A Unified Hierarchical Value Based Oscillatory Network of Location Cells and Path Cells in Avian Brain

A Project Report

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THESIS CERTIFICATE

This is to certify that the thesis titled **A Unified Hierarchical Value Based Oscillatory Network of Location Cells and Path Cells in Avian Brain**, submitted by **Bharat K. Patil**, to the Indian Institute of Technology, Madras, for the award of the degree of **Dual Degree (B.Tech & M.Tech)**, is a bona fide record of the research work done by him under my supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

KEYWORDS: Location Cells; Path Cells; Value Function; Avian Brain.

After the discovery of place cells and grid cells in the Amniote brain, the field of spatial navigation has attracted a lot of attention. Such studies have also led to the discovery of spatial cells in the avian brain. These cells are the building blocks of avian 3-D navigation. This paper propose a modelling approach for *Location Cells* and *Path cells* using Velocity driven value based Oscillatory Network. This model is biologically plausible and shows emergence of place cells and grid cells using only velocity and head direction as input.

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ABBREVIATIONS

IITM	Indian Institute of Technology, Madras
LAHN	Lateral Anti-Hebbian Network
HD	Head Direction
PI	Path Integration
VDON	Velocity Driven Oscillatory Network
PCA	Principal Component Analysis
TD	Temporal Difference
GEN	Go-Explore-NoGo
PID	Partial-Integral-Derivative
UAVs	Unmanned Aerial Vehicles
HF	Hippocampus Formation

NOTATION

θ	Pointing direction of animal's head
θ_i	Preffered direction of i_{th} Head direction neuron
f_0	Base frequency of PI oscillator
β	PI frequency modulation factor
s	Speed of agent/animal
t	time
ϵ_{PI}	PI threshold
Ψ_i	the response of $i_t h$ LAHN neuron
j	Input pattern to LAHN layer
q_{ij}	Forward Hebbian connection in LAHN layer
w_{ik}	Lateral Anti-Hebbian Connections in LAHN layer
η_F, n_L	LAHN layer learning rates
$R(t)$	Reward Signal at any time, t
w_{ij}^I	Weight of MLP of I_{th} layer
δ_i^I	MLP error correction term for I_{th} layer
$\delta(t)$	Temporal Difference Error
$V(t)$	Value at time, t
γ	Discounting Factor
$S(t)$	State vector at time, t
T_{hi}	Higher Value Threshold for GEN
T_{lo}	Lower value Threshold for GEN
σ	Scaling factor for GEN's random update
δ_v	Value Gradient
$\Delta Z(t)$	State update of GEN
ϕ	random update for GEN
T_{lo}	Lower value Threshold for GEN

CHAPTER 1

INTRODUCTION

One of the most important breakthroughs in human history was the invention of aircraft. 100 years later, the current innovation in Aerospace industry is focused on drone navigation, albeit navigation of UAVs (Unmanned Aerial Vehicles) and constructing algorithms and architectures for autonomous navigation.

The core of any autonomous flight architecture involves control and predictive path planning algorithm. The control module in most of the drone technology is currently handled using a PID($Proportional - Integral - Derivative$) module, which works quite fine stabilizing the drones. But the path planning and obstacle hindrance algorithms are mostly based out of conventional GPS tracking. Currently new advancements have been made with the help of Deep Learning models. But these methods are time consuming and lacks the flexibility required to match the performance of natural flying creatures, birds. Maneuvers which are difficult for drones to execute are easily executed by birds during foraging. They achieve this with the help of certain specialised cells in the avian brain.

Just like humans took the inspiration to fly from bird and designed our flights learning from our mistakes over time. It's time we take inspiration for our avian brain from birds too. Thus, with the advent of better computational power it becomes essential to understand the spatial encoding of avian brains to expand our drones' navigational capabilities.

1.1 Background

Spatial navigation means the capability of any organism to move around in its environment. The encoding of space in the brain is an important field of study. The very first idea of spatial maps in the brain were given by Edward Tolman(Tolman, 1948) and their existence were later proved experimentally by O'Keefe and Dostrovsky(O'Keefe

and Dostrovsky, 1971). Later a swarm of neurons have been discovered which altogether form the spatial map in a mammalian brain.

These neurons are called *spatial cells* as they encode the spatial information in the brain. One of the first ones to be discovered had a activation confined to a small space in the environment. These were called place cells because of their space specific firing(O'Keefe and Dostrovsky, 1971). The areas where the neurons are activated are called their *firing fields*. After the discovery of place cells a new class of neurons were reported whose firing dependent only the direction animal was heading(Taube *et al.*, 1990) and not one the position. Such cells were later named *Head Direction Cells*. Another class of cells were discovered later which displayed multiple firing fields, but unlike place cells had a regular hexagonal pattern(Hafting *et al.*, 2005). After this, many neurons like *border cells*(Solstad *et al.*, 2008), *view cells*(Rolls *et al.*, 1997), *speed cells*(Kropff *et al.*, 2015), object cells and object vector cells(Høydal *et al.*, 2019). These cells were found in specific part of the brain called as *Hippocampus-Entorhinal(EC) complex*. All of these cells together form a natural GPS in a mammalian brain that helps the animal forage in an environment.

1.1.1 Avian Brain

All the experiments mentioned above are done on mammalian brains. But to understand navigation for flying objects, we need to understand the spatial encoding aspects of more primitive brains of birds. Birds are known for two different behaviours, migration and homing(Bingman *et al.*, 2006). The first one is movement between two different places and one is the ability to return to the same place again and again. In case of mammals, spatial encoding required for flight is done in bats(Yartsev and Ulanovsky, 2013). But bats have a well developed *Hippocampus-Entorhinal(EC) complex* just like the mammalian brain and place cells and grid cells were reported(Yartsev and Ulanovsky, 2013). But in case of birds, there is no developed hippocampus-EC complex and analogous part of it.

Neural Correlates of Avian Brain

Experiments were conducted on pigeons(*Columba Livia*) with implants in nidostriatum(*hippocampus analogous*) part of the brain. Pigeons were allowed to run in different environments with stable(plus maze: Siegel *et al.* (2005); radial arm: Hough and Bingman (2004)) and unstable goal locations(Kahn *et al.*, 2008). All these reported an emergence of spatially selective cells

Plus maze

Pigeons were made to visit four arms of a plus maze with reward location at the end of the maze(Siegel *et al.*, 2005). There was predictable pattern seen in the way pigeons attended the bowls. The cells showed increased location-correlated activity near the bowls. Most of these cells were later categorized as Location cells(Hough and Bingman, 2004).

Radial arm maze

Like the plus maze pigeons were made to freely navigate in a radial arena(Hough and Bingman, 2004). The activity of isolated HF(*Hippocampus Formation*)-neurons was recorded. The pigeons were reported to take an anticlockwise or a clockwise pattern to visit the bowls. The cell firing from all these neurons was analysed and then categorized in three categories. First, one with a very high spatial selectivity were called ***Location cells***. Their activity was high in confined areas, often near the rewarding locations. Secondly, another class of cells were reported that had a higher firing rate when the pigeon moved between a subset of rewarding points namely ***Path cells***. These cells showed an increased firing rate on the path taken, but they had a drop in their firing rates as they closed in to the rewarding location. Another class of cells were reported namely, ***Arena-off cells***. These cells had a higher firing rates outside the maze in the waiting location but no increased firing was observed in the maze.

CHAPTER 2

BACKGROUND WORK

There are many computational models developed for the emergence of place cells and grid cells. One of the simple plausible models is VDON*(Velocity Driven Oscillatory Network)* model. The model consists of four layers given as HD*(Heading Direction)* layer, PI*(Path Integration)* layer, two LAHN*(Lateral Anti-Hebbian Network)* layers. A random foraging algorithm is used to create a trajectory from which the speed and direction information is calculated. These are the only inputs given to the model.

2.1 Velocity Driven Oscillatory Network

2.1.1 Head Direction layer

It is an array of neurons which is trained as a SOM*(Self Organizing Map)* to give out an activation in the preferred direction of neuron. The output of ith HD neuron is given as the projection of the current direction of the neuron to its preferred direction.

$$\text{HD}_i = \cos\left(\theta - \theta_i\right) \tag{2.1}$$

where, θ_i is the preferred direction of the neuron and θ is the current direction

2.1.2 Path Integration Layer

It is an array of sinusoidal neurons that have one to one connection to the HD neurons. Each sinusoidal neuron is frequency modulated with the current speed(s) and current Heading Direction output of the corresponding HD neuron. Frequency modulation response of ith PI neuron is given as

$$\text{PI}_i = \sin\left[\int 2\pi(f_0 + \beta s \text{HD}_i)dt\right] \tag{2.2}$$

here, β is the modulation factor, HD_i is the Heading Direction output, f_0 is the base frequency of the oscillator. Threshold-ed response of i_{th} PI neuron is given by

$$\text{PI}_i^{th} = \text{H}(\text{PI}_i - \epsilon_{PI}).PI_i \tag{2.3}$$

here, H is given as the Heaviside function and ϵ_{PI} is threshold value.

2.1.3 Lateral Anti-Hebbian Network

Lateral Anti-Hebbian Network is an unsupervised learning neural network that is used for sparse feature extraction from the data set (Földiak, 1990). This behaves like a PCA(*Principal Component Analysis*) over the data set. The network has feed-forward Hebbian and lateral Anti-Hebbian connections. Response of LAHN layer is given as

$$\Psi_i(t) = \sum_{j=1}^{m} q_{ij} X_j(t) + \sum_{k=1}^{n} w_{ik} \Psi_k(t-1) \tag{2.4}$$

Ψ_i is the response of ith LAHN neuron, X is the input patterns from the data set, q_{ij} is the forward Hebbian connection, w_{ik} are lateral Anti-Hebbian connections. m is the input dimension and n is number of LAHN neurons. The weight update rules for feed forward and Lateral Connections is given as

$$\begin{aligned} \Delta w_{ik} &= -\eta_L \Psi_i(t) \Psi_k(t-1) \\ \Delta q_{ij} &= \eta_F [X_i(t) \Psi_i(t) - q_{ij} \Psi_i^2(t)] \end{aligned} \tag{2.5}$$

η_L and η_F respectively are the learning rates. The training is done until the ratio of mutual information of the network to the mutual information of PCA reaches a set threshold.

Table 2.1: Model parameter for the training of VDON model with two LAHN layers

Model Parameters			
PI threshold	0.9	HD cells(N_{HD})	100
LAHN1 threshold	0.75*max	LAHN1 cells(N_{L1})	64
LAHN2 threshold	0.9*max	LAHN2 cells(N_{L1})	48
dt	0.01s	β	70
η_L	0.01	η_F	0.01

There are two LAHN layers in the network, the first one serves as an input to the next one. The firing pattern of first layer filtered over a threshold resembles that of a hexagonal grid cells. The second LAHN layer when filtered over a threshold resembles the firing pattern of place cells. The emergence of place cells in LAHN 2 signifies that the neurons had encoded spatial information. Later many experiments were carried out with multi-compartment environments(Jayakumar *et al.*, 2019), Saccade velocity(Chauhan *et al.*, 2019) etc.

CHAPTER 3

Value Function Based Foraging Network

This chapter focuses on the modelling of a value based navigation network which trains for value space over a rewarding environment. The input of the model is LAHN 2 output without a threshold. This model is then tested using a hill climbing algorithm to generate a self-directed trajectory by the agent.

3.1 Oscillatory Value Function Model

3.1.1 Spatial Encoding in LAHN 2 layer

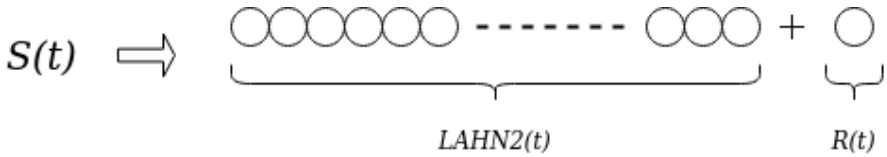
Before modelling the value space, a check was performed on LAHN 2 output for the retention of spatial information. This was formulated as a regression problem using a fully connected network. An MLP($MultilayeredPerceptron$) was trained with one hidden layer and two output nodes in the output layer. The LAHN 2 response at time t was given as an input pattern and was trained against the actual position of agent at time t. Accuracy of the model suggested that decent spatial information was retained in the LAHN 2 output for t. Accuracy could have been increased more by parameter tuning but that was not the motive of the test.

This test was a checkpoint for validity of all the data generated using VDON model. All trajectories and data generated for training the value model was passed through this. In case of lesser accuracy than 90%, VDON model was retrained with different parameter values.

3.1.2 Value Space

Value for any state in a state space is given as the expectation of total reward for that state. In case of our model, each individual state $S(t)$ at any time, t, is given as an (N+1) dimension vector, where N is the number of neurons in the LAHN 2 layer. The

last node gives the reward signal which is denoted as $R(t)$ at any time, t.



The value space, $V(S)$, is generated over the state space using a fully connect network(MLP) with one hidden layer and single node in output layer. The network was trained using an unsupervised reinforcement learning algorithm called Temporal Difference(TD) Learning. Weight and bias term update for the fully connected network is given by

$$\begin{aligned} \Delta w_{ij}^{I} &= \eta \delta_i^I V_j^{I-1} \\ \Delta \theta_i^I &= -\eta \delta_i^I \end{aligned} \tag{3.1}$$

where, η is the learning rate, w_{ij} is the weight between i^{th} neuron in I^{th} layer and j^{th} neuron in $(I-1)^{th}$ layer. V^{I-1} is the output of $(I-1)^{th}$ layer, which comes as an input to I^{th} layer. The error correction, δ_i^I is calculated using the back propagation algorithm. The error term used for the output layer is called the Temporal Difference(TD) error, δ and is given as

$$\delta(t) = R(t) + \gamma V(t+1) - V(t) \tag{3.2}$$

$\delta(t)$ is the TD error, $R(t)$ is the reward signal, $V(S)$ is the value function at time,(t), calculated by giving state, $S(t)$ as an input to the network and γ is the discount factor.

3.1.3 Training and Setup

The training of the model is done in two stages with two different trajectories. The trajectories differ in their presentation of reward signal, $R(t)$.

Environments

Two different environments are designed for the experiment. First one is a square box of side, a. Four circular areas tangent to all four edges are called bowl areas and are designated a specific reward.

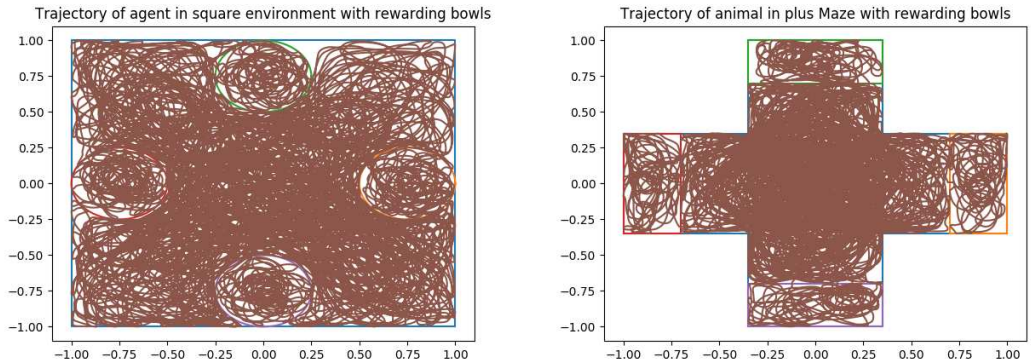


Figure 3.1: Both the environments with second trajectory type

Table 3.1: Environment dimensions and Setup parameters

Plus maze			
Arm length, l	2	Arm width, b	0.7
Bowl length, b	0.7	Bowl width, bb	0.3
Reward $x - axis$, $R1$	1	Reward $y - axis$, $R2$	0.5
Least time in bowl, $t1$	80s	Reward presentation, T	55s
Box Environment			
Side length, a	2	Bowl radius, r	0.3
Reward $x - axis$, R_1	1	Reward $y - axis$, R_2	0.5
Least time in bowl, t_1	100s	Reward presentation, T	75s

Second environment is a plus maze with the rectangular bowl areas of width, bb and length, b at the ends of each arm. The width and length of each arm is b and l respectively. Same plus maze was used in the homing pigeon experiments(Siegel *et al.*, 2005), to see goal specific *Location Cells*

Trajectories and Reward

For both the environments two different trajectories were generated. First one is a normal free programmed trajectory for an environment. Free here means that there were no boundaries for the bowls and the agent was free to move everywhere. The points that were contained in the bowl area were sampled out and were designated a reward of R_1 for the bowl areas on *x-axis* and R_2 for the bowl areas on *y-axis* 3.1. The reward was presented for the entire time the agent moved inside the bowl area.

The second trajectory assumed bowls to have boundaries. As the animal came in ϵ vicinity of the any of the bowls, it could enter the bowl. It had to spend at least

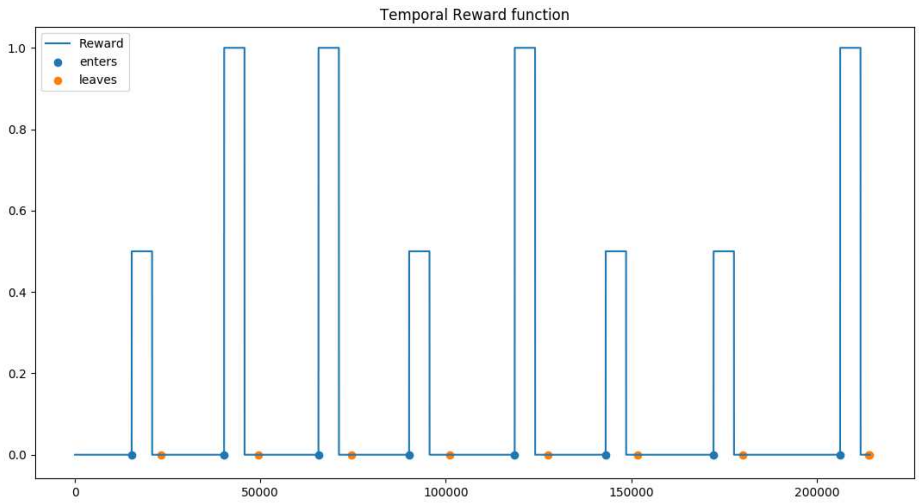


Figure 3.2: Reward signal presented for second trajectory. The reward is presented only for time T

minimum time(t_1), foraging inside the bowl before it could move out. During this t_1 time, the reward signal was presented for time(T), where $T < t_1$, irrespective of the bowl it was in. The reward values, R_1 and R_2 stayed the same. This means that the moment agent entered a bowl, it started getting a reward which then drops abruptly after T time, even if the animal is still in the bowl. This is plausible with the real experiments as the food pellets are finite for any bowl.

Network Architecture and Training

Network consists of an input layer, one hidden layer and an output layer. Input layer has same dimensions as the state vector $S(t)$ and output layer has a single node to output the value function $V(S)$. The dimension of hidden layer were kept same as the dimension of network in section 3.1.1 to maintain consistency in the results.

The network was trained first for the normal free trajectory with states, $S_1(t)$. The weights and biases were saved after the loss reached a specific threshold. These weights were then further trained with states, $S_2(t)$ from the second trajectory which had a temporal reward signal. Two different models were trained for both the environments separately.

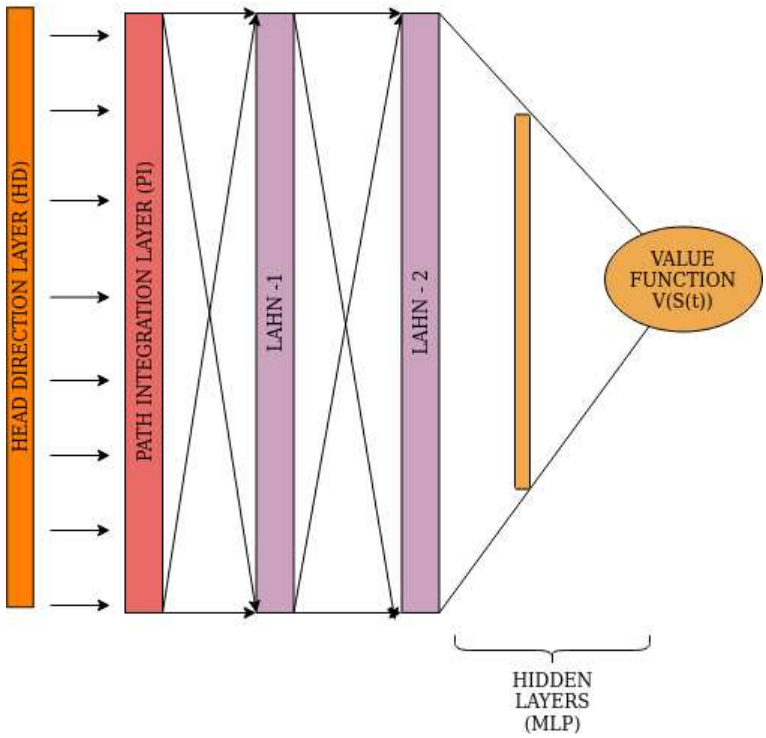


Figure 3.3: Full Value model architecture with VDON.

Table 3.2: Network architecture and Gen algorithm Parameters

Value model parameters			
State dimensions N	49	hidden layers	1
Hidden layer neurons	200	Discount factor, γ	0.5
Learning rate, η	0.001		
GEN parameters			
High Value Threshold, T_{hi}	0.7	Low Value Threshold T_{lo}	-0.7
σ	0.35		

3.2 Testing

The trained model gave Value, $V(S)$ as an output when the state, $S(t)$ was given as an input. For testing we needed the animal to forage the environment based on the value it sees. Final goal is to see an analogous clockwise or anticlockwise trajectory as taken by birds in referenced experiment(Siegel *et al.*, 2005). For this value function based state prediction algorithm, GEN($Go - Explore - NoGO$) was used(Chakravarthy and Moustafa, 2018). GEN has been shown as a effective hill climbing algorithm for value function based computational models of the Basal Ganglia.

GEN*(Go-Explore-NoGo)*

GEN is a hill climbing algorithm which predicts the next state of the model $Z(t+1)$ on the basis of the value gradient, δ_v, at current time, t. The algorithm works in three regimes, (a) Go regime for large positive δ_v (b) Explore regime for intermediate δ_v and, (c) No-Go regime for large negative δ_v. The update for the next state $Z(t+1)$ is given as

$$\begin{aligned} &if(\delta_v > T_{hi}) \\ &\qquad \Delta Z(t+1) = \Delta Z(t) \\ &elseif((\delta_v > T_{lo})\&(\delta_v < T_{hi})) \\ &\qquad \Delta Z(t+1) = \phi \\ &elseif(\delta_v \leq T_{lo}) \\ &\qquad \Delta Z(t+1) = -Z(t) \end{aligned} \tag{3.3}$$

Here,

- $\delta_v = V(t) - V(t-1)$
- $\Delta Z = Z(t) - Z(t-1)$
- $V(Z)$ = Value function
- $Z(t)$ = state vector of N dimension

Here, T_{ho} and T_{lo} are high and low thresholds for Value function respectively. The interpretation of above algorithm is such that, if the value gradient δ_v is higher than a positive threshold T_{ho} that mean it repeats the last update. For a δ_v value smaller than a negative threshold T_{lo}, the update send the agent back to older state. For an intermediate δ_v, that is if it falls between T_{ho} and T_{lo}, the agent starts exploring. Hence a random update, ϕ, is done to the state, which is given as

$$\phi = G(0,1)\exp\left[\frac{-\delta_v^2}{\sigma^2}\right] \tag{3.4}$$

Here, $G(0,1)$ is a random Gaussian variable with mean zero and standard deviation 1, $\exp\left[-\delta_v^2/\sigma^2\right]$ is the scaling factor.

In value network, state vector, $S(t)$ is given as the input to the model but GEN update

is given directly to the position co-ordinates. The GEN update is not done to the $S(t)$ vector, because of its dimensionality. Rather, it's done to the position vector as the position information is anyway getting encoded in the $S(t)$ vector alongside Reward, $R(t)$.

Initial position and initial step is randomly selected for the agent and then the agent is left to forage the environment on its own.

CHAPTER 4

Results and Conclusion

The model was trained and tested with the reward scheme as given in Table 3.1. The hidden neurons in the MLP were expected to show goal specific activity like *Location cells*.

4.1 Results

The heat maps shown in Figure 4.4 and Figure 4.2, show that for the bowls having higher rewards the value function had higher values. Also when the rewards dropped the value function dropped drastically. The firing pattern shown for the hidden layers neurons in Figure 4.4 and Figure 4.2, shows that the hidden neurons are in fact encoding the position of goal areas. There needs to be a second training without the inclusion of $R(t)$ in state $S(t)$.

4.1.1 Plus Maze

Value Function and hidden layer neurons

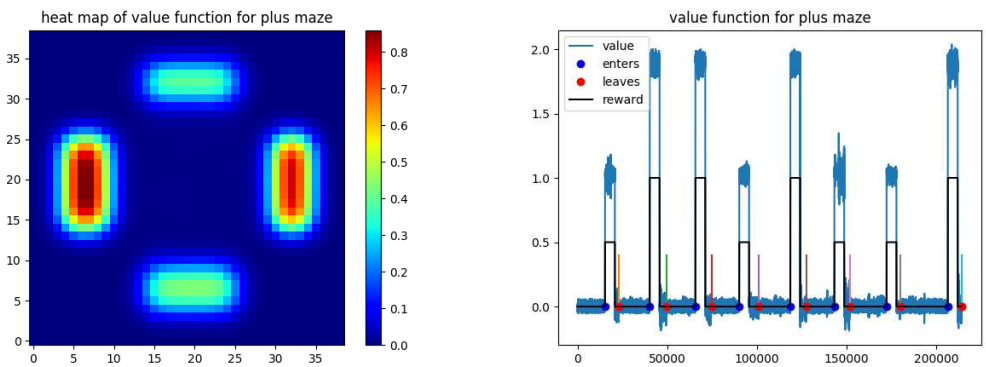


Figure 4.1: Value space over plus maze shows higher value at points with higher reward(left). Value function shows a strong dependence on reward(right)

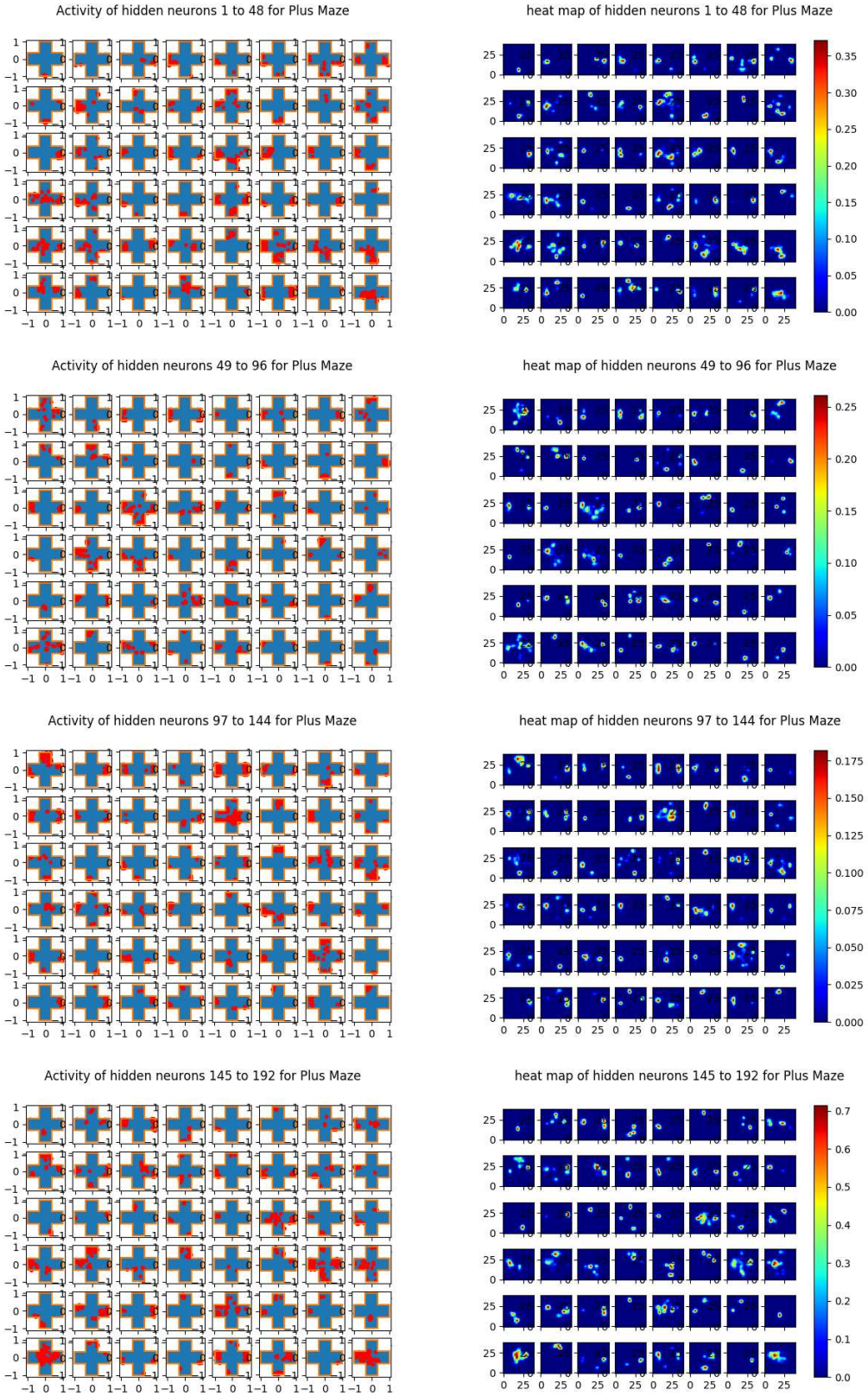


Figure 4.2: Firing Pattern of hidden layer neurons in model trained over Plus maze.

4.1.2 Box Environment

Value Function and Hidden layer neurons

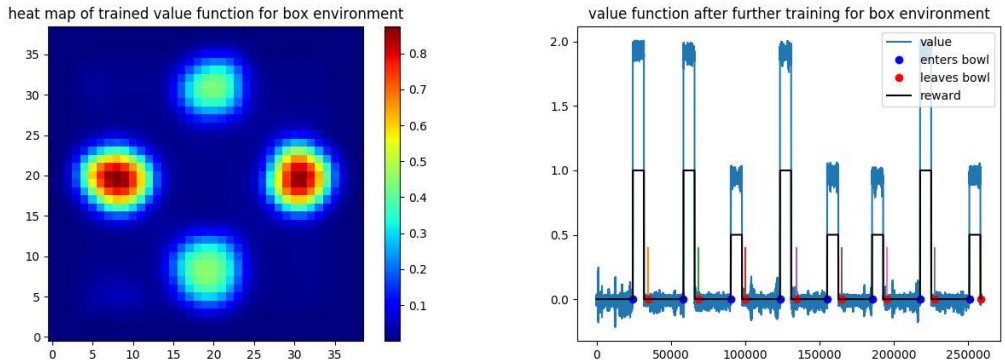
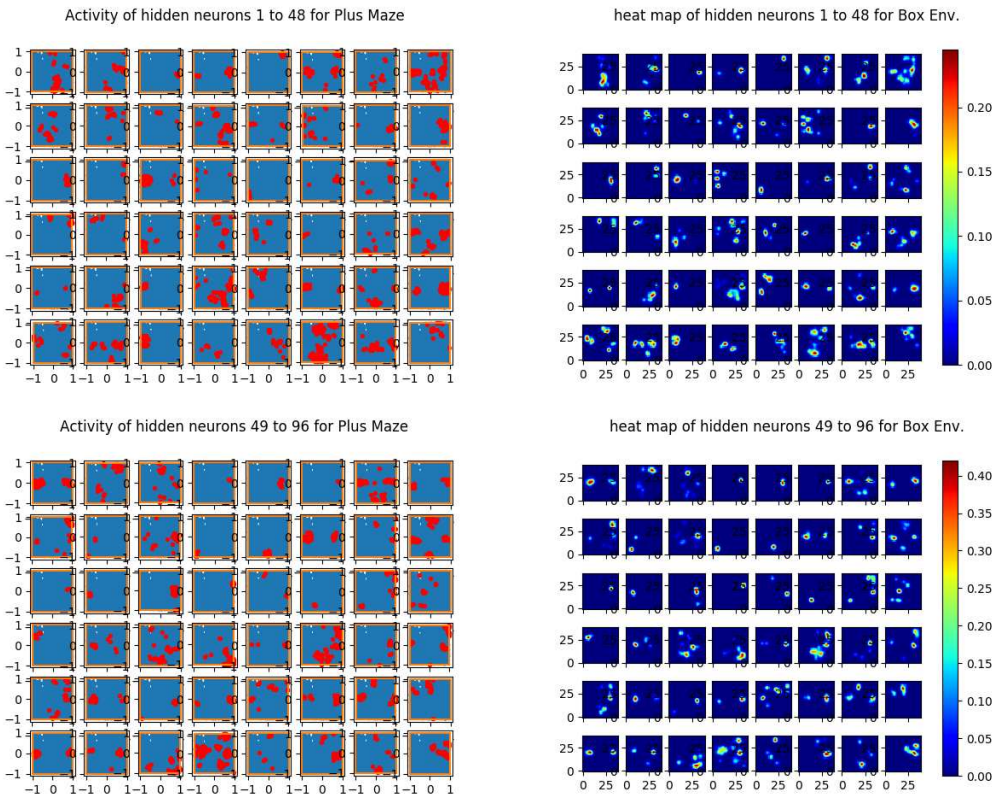


Figure 4.3: Value space over box env. shows higher value at points with higher reward(left). Value function shows a strong dependence on reward(right)

Hidden layer neurons and Location cells



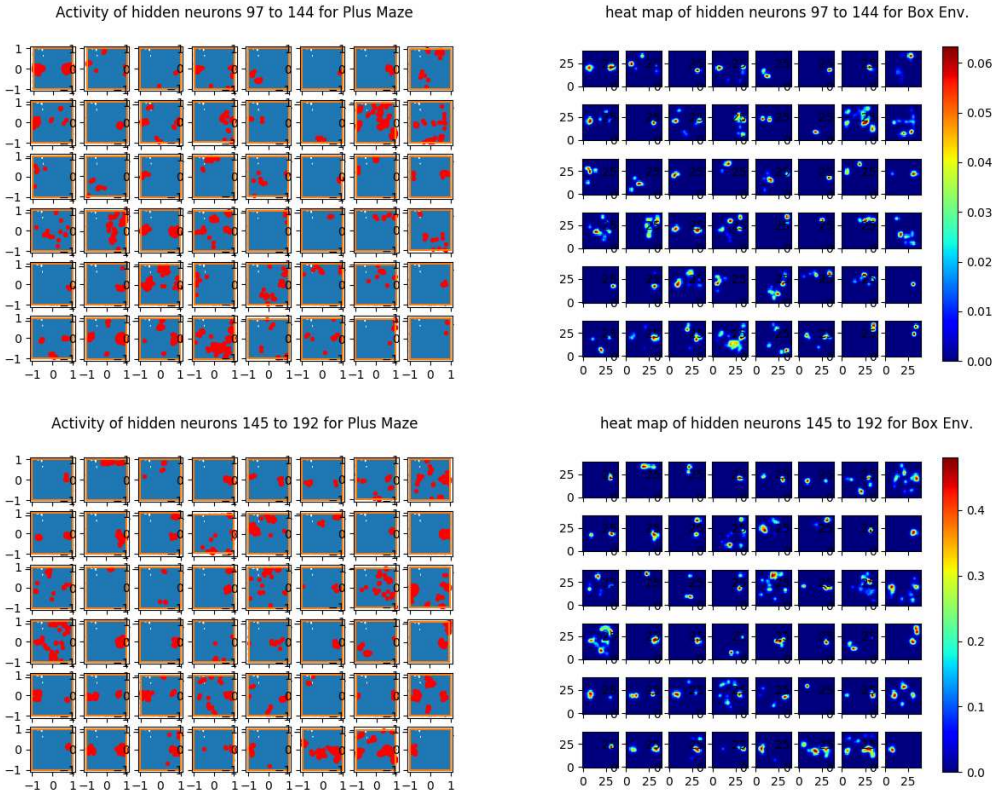


Figure 4.4: Firing Pattern of hidden layer neurons in model trained over Box Environment.

4.2 Conclusion

- A fully connected layer trained with TD error is able to capture reward based spatial information
- Nearly 70-75% of hidden layer neurons showed firing near the goal/rewarding locations.
- Reward $R(t)$ needs to be removed from the input state $S(t)$, as it is presented in the form of δ_v to the model.
- There was no predictable pattern observed to visit the bowls, in both the environments using GEN algorithm.
- There seems to be a need of Actor network for more directed navigation.
- There is a need for more experiments to give a conclusive remark on the emergence of spatial cells in the model which are analogous to the avian brain

CHAPTER 5

Future Work

It is evident from the results that the value model is able to replicate the all the reported results of the *Location Cells*. Another class of cells called path cells(Hough and Bingman, 2004) are reported in avian brain. These cells had an increased activity when the pigeon moved between a subset of rewarding points. The firing dropped as the pigeon closed in near the rewarding bowls. The model was not able to exhibit such cells. The reason being that the above explained model is from a set of Reinforcement Learning models called as *critic networks*.

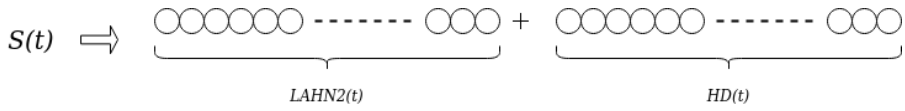
Actor-Critic networks

Actor-critic models are the foundation of any modern RL(Reinforcement Learning) algorithms. There are two method of RL; value based and policy based. Actor critic is a merge of both. Main idea is to train two networks rather than one. One model is used to generate *Q-values*,value for every action at every state(critic network). Another model is responsible for outputting the action based on state(actor network). In above model, an actor network can be trained to give out action one the basis of state. $S(t)$.

Head direction concatenation

The input to the model is the state vector, $S(t)$, which comprises of the LAHN 2 output and the Reward, $R(t)$. The value is calculated on the basis of this state. LAHN 2 has enough spatial information, but only position does not define state in our context. Because an animal standing at some position and heading left is in a different state than the animal standing at the same position but heading right. A heading information needs to be added to the state vector,$S(t)$. The head direction response from the VDON model will be concatenated to the state vector, $S(t)$, to give a complete state of the animal. Also, the reward as an entity is given to the network in form of TD error, and hence the reward scheme should not be given as an explicit input to the model. Finally,

the state of the animal is defined by LAHN 2 response and HD response concatenated. This in turn will be given as an input to the actor as well as the critic model.



Further

Another experiments need to be done in order to look for location cells in radial arm maze(Hough and Bingman, 2004). The stability of the location cells using visual cues(Hough and Bingman, 2008). Emergence of Avian spatial cells in unstable goal settings(Kahn *et al.*, 2008). There are many experiments that need to be done in the project and start understanding the path planning capabilities of birds.

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